

7.2. a

označme bázi: $K = \{1, x, x^2\}$

$$[f]_K^K = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix} \text{ hčďdane char. pol.}$$

$$P_f = \lambda^2(2-\lambda) \quad \text{v.l. č. } 2 \quad \text{a } 0$$

A.h. 1 a.h. 1

$$M_2 = \text{Ker} \begin{pmatrix} -2 & 2 & 0 \\ 0 & -2 & 4 \\ 0 & 0 & 0 \end{pmatrix} = \text{Span} \left\{ \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \right\} \quad b_1 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}_K$$

$$M_0 = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\} \quad b_2 = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}_K$$

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}_K \mapsto \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}_K \mapsto \vec{0} \quad b_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}_K$$

tedy báze $(2+2x+x^2, 2, x) = B$ splňuje

$$[f]_B^B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$